BECOME A KAGGLE MASTER - HW1

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Sommaire

Exploratory data analysis - 1

Quantiles, unique values and types

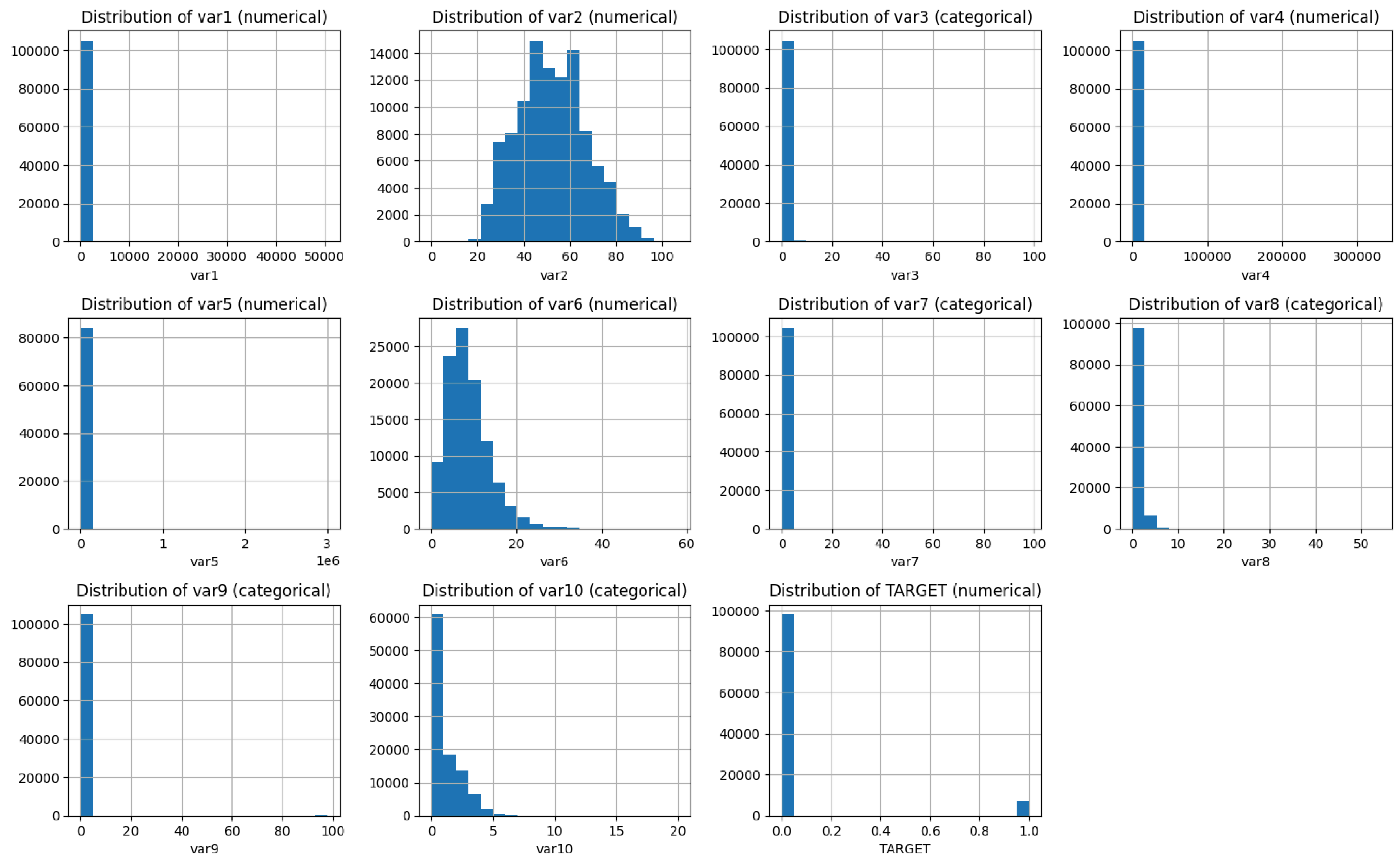
	mean	std	min	25%	50%	75%	max	unique_values	type
var1	6.073156	257.859625	0.0	0.030002	0.155936	0.561917	50708.0	88523	float64
var2	52.261014	14.768987	0.0	41.000000	52.000000	63.000000	107.0	85	int64
var3	0.422665	4.221443	0.0	0.000000	0.000000	0.000000	98.0	15	int64
var4	354.627892	2104.703046	0.0	0.175067	0.366162	0.866584	329664.0	82416	float64
var5	6636.738727	13594.929730	0.0	3400.000000	5400.000000	8228.000000	3008750.0	12038	float64
var6	8.442294	5.136260	0.0	5.000000	8.000000	11.000000	58.0	57	int64
var7	0.268386	4.197747	0.0	0.000000	0.000000	0.000000	98.0	18	int64
var8	1.018696	1.134364	0.0	0.000000	1.000000	2.000000	54.0	28	int64
var9	0.242186	4.184389	0.0	0.000000	0.000000	0.000000	98.0	13	int64
var10	0.757957	1.116196	0.0	0.000000	0.000000	1.000000	20.0	12	float64

Categorical variables already encoded

Distribution of target

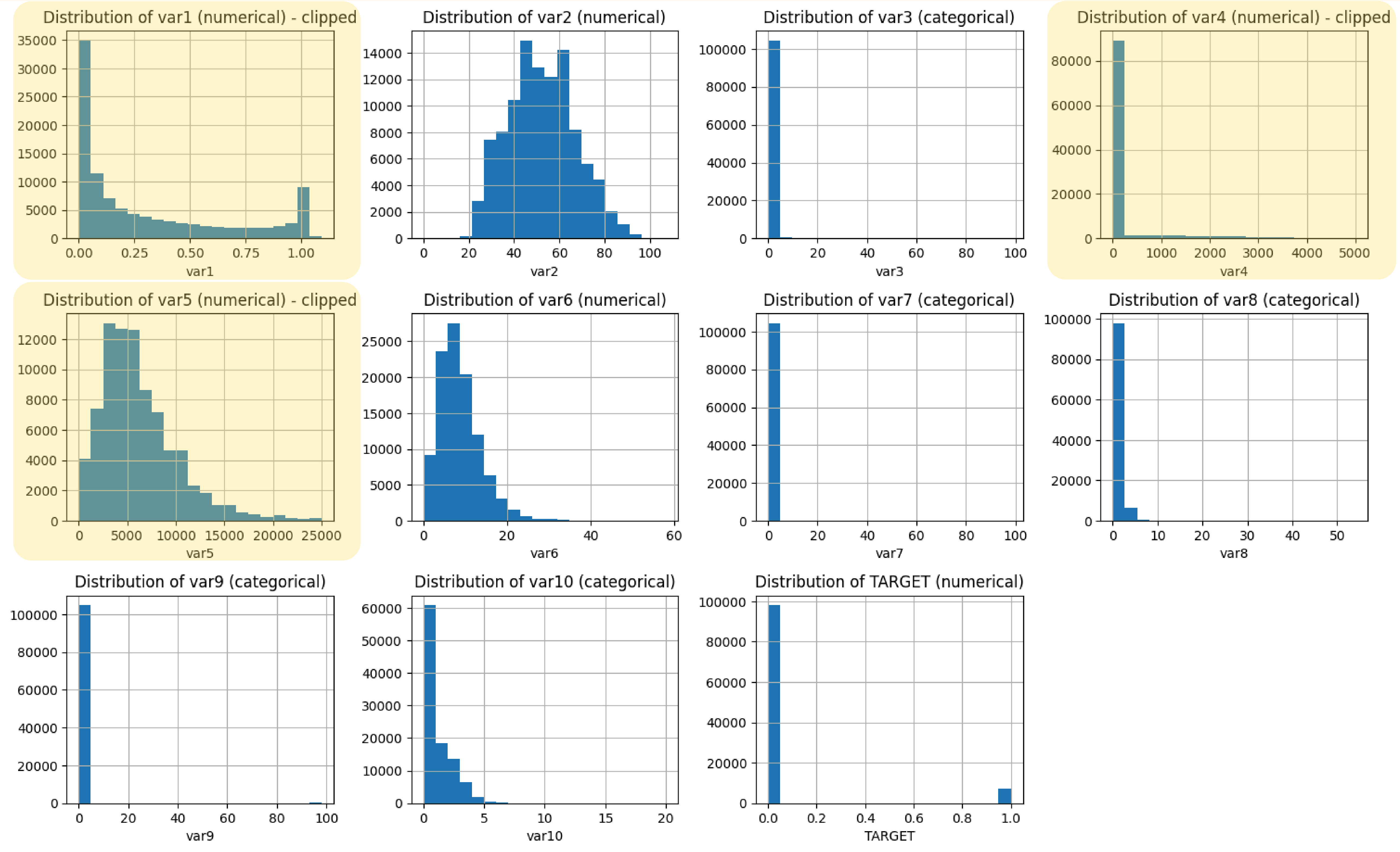
93.3% of class 0 - 6.7% of classe 1 → unbalanced

Exploratory data analysis - 2



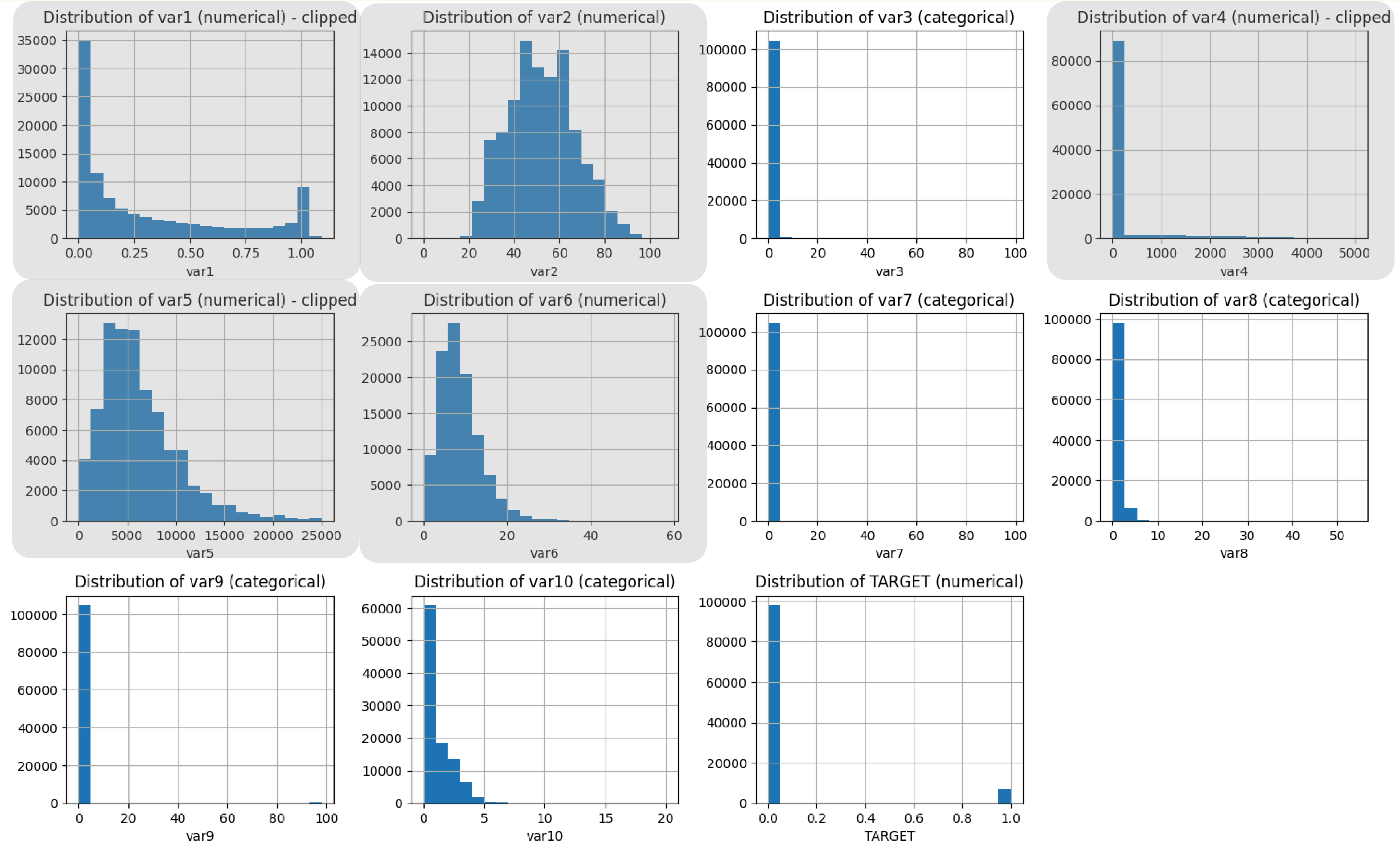
Exploratory data analysis - 3

Extreme values : clip at 99th centile



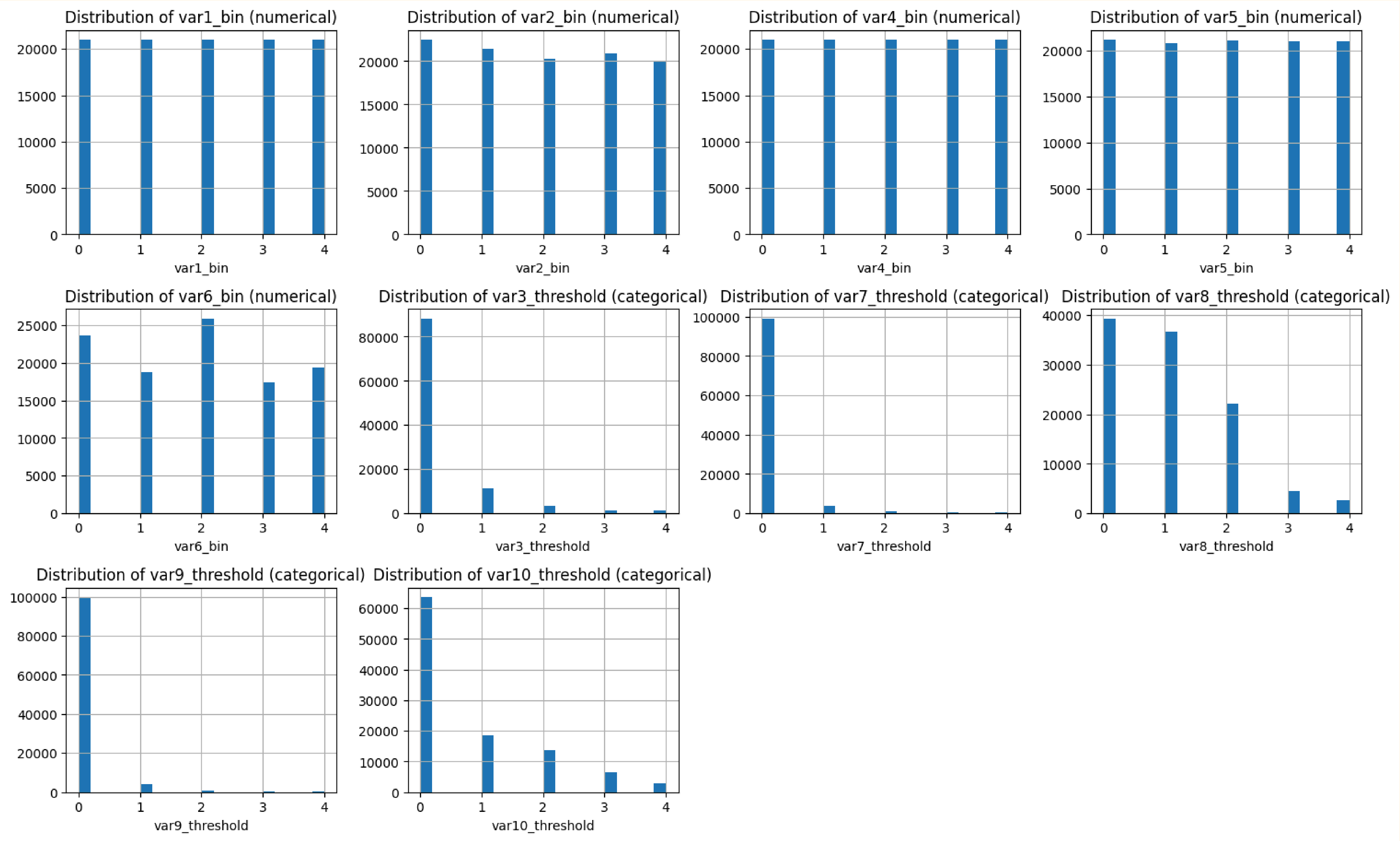
Exploratory data analysis - 4

Log transformation ; Box-Cox transformation



Exploratory data analysis - 5

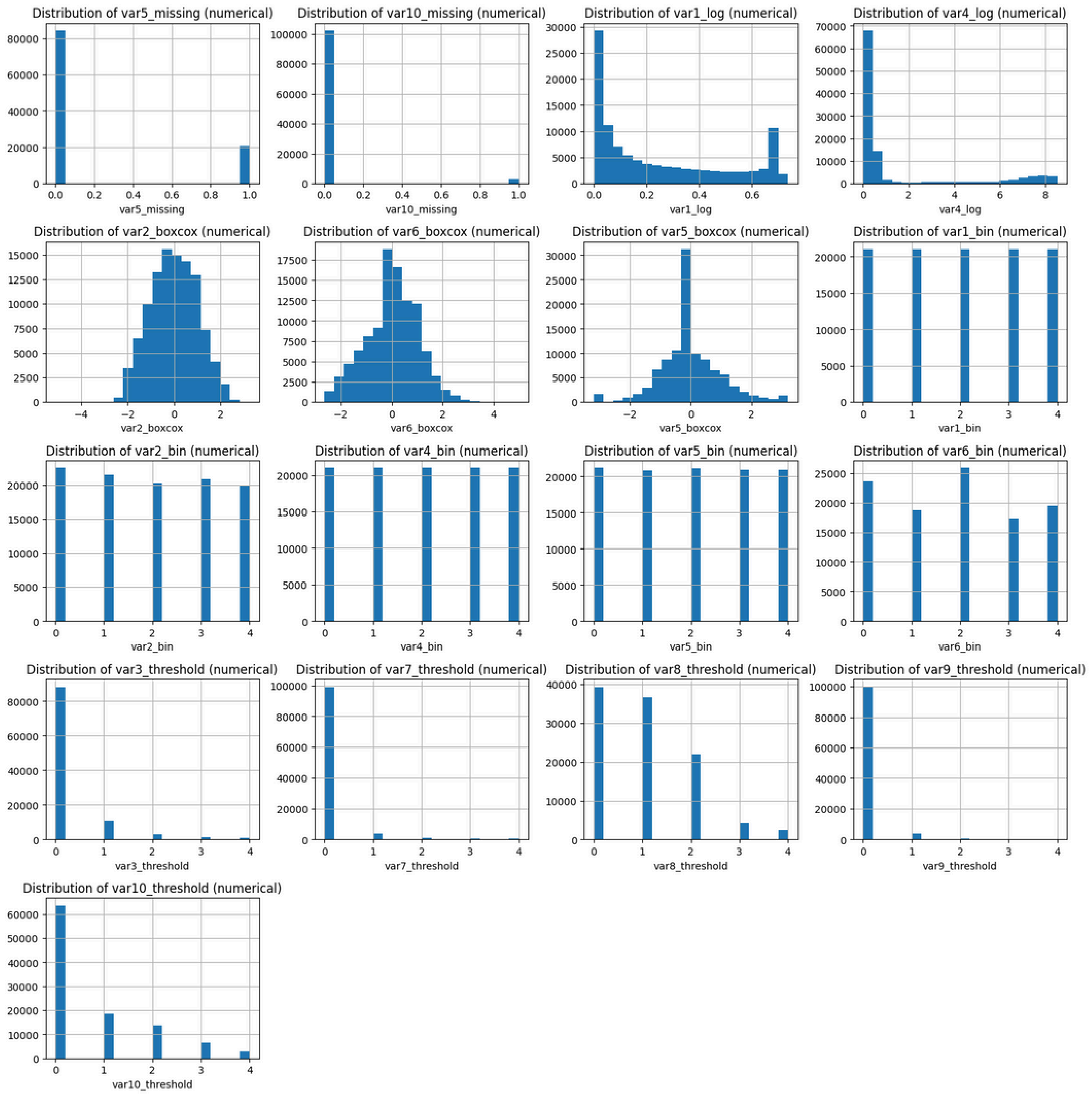
Binning - by quartile or by threshold



Exploratory data analysis - 6

Final features:

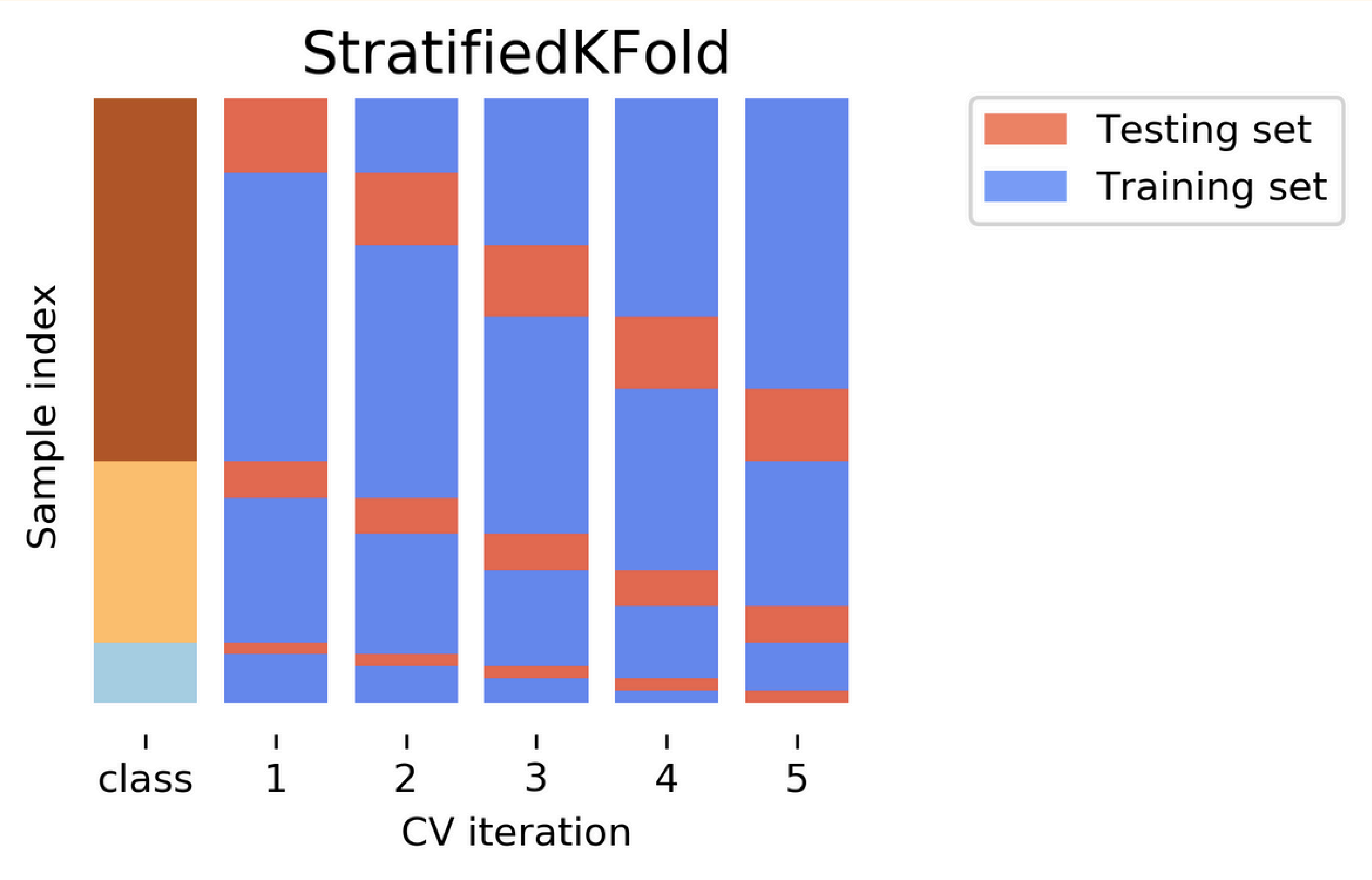
- Delete initial columns
- 17 features kept
- var3, var7 et var9 with thresholds are not as correlated as before
- Missing values replaced with median



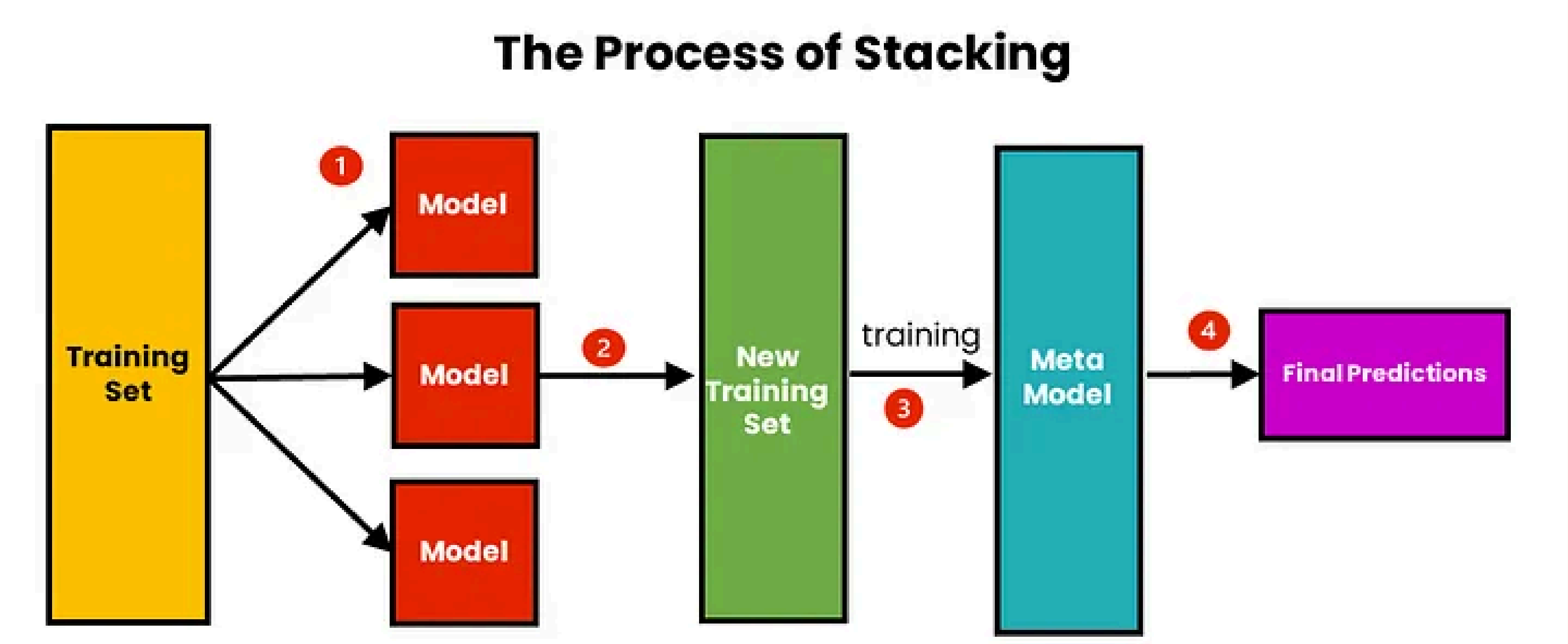
K-Fold validation

StratifiedKFold with k = 5 :

- Ensures an equal target classes distribution
- Check if no overfitting



The Process of Stacking



Objective:

- Train a metal-model to optimally combine the predictions of base models

Pros:

- Can reduce bias and variance
- It combines the strengths of several models

Cons:

- Risk of overfitting
  - Solution: k-fold Cross Validation
- Increase the complexity

Model Selection : Base Models

1) CatBoost Classifier:

- Best algorithms for **structured/tabular data**
- handle imbalance
- less prone to overfitting thanks to regularization
- handle categorial features well

2) Logistic Regression

- Simple: detect linearly separable patterns

3) MLP Classifier:

Neural Network

- Nonlinear relationships: can find patterns not seen by the tree algorithms

4) XGBoost Classifier:

- Extremely powerful on **tabular** and **imbalanced** data
- Widely used in ML

Objective:

- Balanced Complexity
- Imbalance Handling
- Diversity: Linear, Tree-Based, Neural Network

Model Selection : Meta Model

Objective:

- Combine outputs (predictions) of the base models
- Adjust these predictions to make the final decision

Meta Model:

- Ridge: Linear Regression with Regularization L2

Pros:

- Simple and easy to interpret
- It does not require much hyperparameter tuning
- It limits the impact of overly similar or unreliable models

Cons:

- It does not learn non-linear interactions between models

Use of CalibratedClassifierCV

Objective:

- Adjust the **predicted probabilities** of a classifier in order to better reflect the true likelihood of and event
- If they are not well-calibrated the meta-model might learn incorrect patterns

Example:

- A model predict 0.95 probability for a class but only be correct 75% of the time
  - We adjust it to 0.6

Hyperparameter tuning

1. Initial tuning → fix hyperparameters
2. Improving data processing
3. New tuning with final data processing

   The new hyperparameters improves again the model performances

Result

RMSE on train dataset : 0.2216